

TAIGO SAKAI

PH.D. STUDENT | COMPUTER VISION RESEARCHER

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RESEARCH AREAS

Continual Learning
Long-Tailed Learning
Multi-View Detection & Tracking
Deep Learning for Medical Imaging

TECHNICAL SKILLS

Python / PyTorch
Deep learning research and experimentation
OpenCV / Computer Vision
Detection, tracking, image analysis
Linux / Git / GitHub
Research infrastructure and reproducible development
iOS / Web Development
Prototype and application development

WEB & SOCIAL

Portfolio
GitHub · sakai1250
Google Scholar
LinkedIn · sakai1250

CERTIFICATIONS

JDLA Deep Learning Engineer
Japan Deep Learning Association · 2024
Third-Class Electric Chief Engineer
Japan · 2023

SUMMARY

Ph.D. student at Meijo University working on computer vision and continual learning. Research interests include long-tailed learning, dynamic neural capacity, and robust multi-view pedestrian detection and tracking. Author of peer-reviewed international publications and recipient of multiple research and engineering awards.

EXPERIENCE

Special Assistant

Meijo University · Apr. 2026 - Present

- Supports research and education while pursuing doctoral research in computer vision and deep learning.

Researcher / Graduate Student

Hotta Laboratory, Meijo University · 2023 - Present

- Develops methods for continual learning, class imbalance, and multi-view pedestrian detection and tracking.
- Presents research at international and domestic computer vision conferences and workshops.

EDUCATION

Ph.D. Course, Electrical, Electronic, Information and Materials Engineering

Meijo University Graduate School · Apr. 2026 - Present

Master's Course, Electrical and Electronic Engineering

Meijo University Graduate School · Apr. 2024 - Mar. 2026

B.E., Electrical and Electronic Engineering

Meijo University · Apr. 2020 - Mar. 2024

SELECTED PUBLICATIONS

T. Sakai, K. Hotta

NeuroGuard: Neural Gradient Update Aware of Representation Damage
ECCV 2026 Workshop on Human-inspired Computer Vision · Oral

T. Sakai, K. Hotta

Dynamic Distillation and Gradient Consistency for Robust Long-Tailed Incremental Learning
ICIP 2026 · Poster

T. Sakai, T. Nakamura, H. Shimizu, K. Hotta

Visibility-Gated ConvGRU for Robust Multi-View Pedestrian Detection and Tracking in BEV Space
VISAPP 2026 · Oral

T. Sakai, K. Hotta

Long-Tailed Data Classification by Increasing and Decreasing Neurons During Training
ICCV 2025 Workshop on Human-inspired Computer Vision · Poster

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HIGHLIGHTS

7 peer-reviewed international papers

2024 - 2026

DIA2026 Encouragement Award

3 / 96 presentations

ViEW2025 Algorithm Contest

1st / 131 entries

MIRU2025 Student Encouragement Award

55 / 726 submissions

ACADEMIC SERVICE

Technical Program Committee

ECCV 2026 Human-inspired Computer Vision Workshop

Technical Program Committee

ICCV 2025 Human-inspired Computer Vision Workshop

SELECTED AWARDS

IEICE Tokai Branch

Student Research Encouragement Award · 2026

Meijo University

President's Award · 2023, 2024, 2025

IPSJ CVIM

Best Bachelor's Thesis Session Award · 2024

LINE Yahoo HackU

Grand Prize · 2023 / HappyHacking Award · 2023, 2025

Power Academy

First Prize · 2023 / Excellence Award · 2021

PUBLICATIONS

K. Toida, T. Sakai, T. Nakamura, H. Shimizu, K. Hotta

ATLASFusion: Aggregation Tracking with Location-Aware Sparse Fusion for Robust Spatio-Temporal Multi-View Pedestrian Tracking

ECCV 2026 Workshop on Computer Vision for Multimedia Spatial Intelligence through Time · Poster

T. Sakai, K. Hotta

Improving Noise Robustness in Medical Image Classification via Dynamic Integration of CNN and KAN

CARS 2026 · Poster

T. Sakai, K. Hotta

Growing Deep Neural Network Considering with Similarity between Neurons

ECCV 2024 Workshop on Human-inspired Computer Vision · Poster

SELECTED DOMESTIC PRESENTATIONS

T. Sakai et al.

Gabor-KAN: Medical Image Classification with Complex Gabor Wavelets for Kolmogorov-Arnold Networks

MIRU 2026 · Peer-reviewed poster

T. Sakai et al.

Visibility-Gated ConvGRU for Robust Multi-View Pedestrian Detection and Tracking in BEV Space

DIA 2026 · Peer-reviewed oral

T. Sakai et al.

Gradient Consistency Regularization and Dynamic Knowledge Distillation for Continual Learning on Imbalanced Data

MIRU 2025 · Peer-reviewed oral

T. Sakai et al.

Long-Tailed Data Classification by Increasing and Decreasing Neurons During Training

ViEW 2024 · Peer-reviewed poster

T. Sakai et al.

Growing Deep Neural Network Considering Neuron Similarity Distribution

IPSJ CVIM 2024 · Oral

SELECTED AWARDS & GRANTS

DIA2026 Encouragement Award

Workshop on Dynamic Image Processing and Application · 3/96

ViEW2025 Visual Inspection Algorithm Contest - First Prize

1st place among 131 entries

MIRU2025 Student Encouragement Award

55 selected among 726 submissions

JASSO Scholarship Repayment Exemption for Outstanding Achievement

Half exemption · 2026

Rinnai Scholarship Foundation

Selected scholarship student · 2026

Telecommunications Advancement Foundation

Overseas travel support · 2026